## Independent Position Determining – General Data

1. General

The independent position determining system has the following sub-systems:

A. Weather Radar System

Refer to 34-41-00, Weather Radar System (Mod 6/2049) – General Data for system details.

The RDR 2000 weather radar (Radio Detecting and Ranging) system gives the pilot a selectable horizontal display of thunderstorms or high density precipitation in front of the aircraft. When a pulse intercepts a target, the energy is reflected as an echo, or return signal, back to the antenna. From the antenna, the returned signals are transferred to the receiver/transmitter unit. The returned signals, are then displayed.

For aircraft incorporating customer Standard Order Option S.O.O. 6388 the Weather Radar System is not installed.

For aircraft with S.O.O. 6388 incorporated, the Weather Radar System and associated systems are installed by incorporating customer Standard Order Option S.O.O. 6383.

B. Stormscope System

Refer to 34-41-10, Stormscope – General data for system details.

The Stormscope (WX–500) System maps electromagnetic signals (lightning) 360 degrees around the aircraft and to a maximum radius of 200 nautical miles.

The Stormscope system consists of a receiver installed in the rear avionics rack and a top mounted antenna installed at FS400. A receiver inhibit line is interfaced to the aircraft audio control panel ensuring that erroneous strike signals are not displayed when a radio transmitter is keyed.

C. Radar Altimeter System

Refer to 34-42-00, Radar Altimeter System (Mod 6/2049) – General Data for system details.

The KRA 405B radar altimeter transceiver is installed in the empennage aft of the rear baggage compartment. A second radar altimeter can be fitted if desired. The transceiver sends a signal to the transmit antenna and gets the return signal from the receive antenna. The maximum operating height AGL used by the system is 2,500 ft.

The radar altimeter system measures the aircraft height Above Ground level (AGL) electronically and sends the height AGL data to the Modular Avionics Unit (MAU) for display in the ADI window of the left and right PFDs. The digital readout for radio altitude is displayed in white text in a black box in the lower center part of the attitude display on the PFD. The radar altitude display is removed at altitudes greater than 2,500 ft. When altitude is less than 550 feet, the lower portion of the PFD altitude tape will show a yellow cross hatched box to indicate the ground proximity.

If the radar altitude data becomes invalid, the digital readout will be replaced with RAD in white in an amber box. The radar altimeter data is (in addition to indicated airspeed data from the ADAHRS) is used by the aircraft air-ground monitor logic system to determine on ground or in-flight status.

The radar altimeter is an integral component of the Class A TAWS (Terrain Awareness and Warning System) and must be installed in aircraft that are ordered with TAWS.

D. Traffic Alert and Collision Avoidance Systems

Refer to 34-43-00, Traffic Alert and Collision Avoidance System (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System (TCAS II) (Mod S.O.O. 6219) – General data for system details.

A KTA 910 TCAS I or TPU–67A TCAS II can be fitted. The TCAS display is integrated into the Apex system, and TCAS alerts are voiced through the aural warning system.

Both TCAS systems use a processor, two directional antennas (upper and lower) and a configuration module. Aural alerts are provided through the audio panel and from there onward to the crew headsets.

TCAS I provides traffic advisories (TA), but does not provide resolution advisories (RA). TCAS II provides both traffic advisories (TA) and resolution advisories (RA).

For aircraft with S.O.O. 6388 incorporated. the Traffic Alert and Collision Avoidance System (TCAS I) is not installed.

For aircraft with S.O.O. 6388 incorporated, the Traffic Alert and Collision Avoidance System (TCAS I) and associated systems are installed by incorporating customer Standard Order Option S.O.O. 6382.

E. Enhanced Ground Proximity Warning System (TAWS)

Refer to 34-44-00, Enhance Ground Proximity Warning System (TAWS) (Mod 6/2049) – General Data for system details.

The Enhanced Ground Proximity Warning System (EGPWS) (TAWS) provides alerts and warnings to prevent controlled flight into terrain. This is part of the Public Commercial Operation Bundle customer option order 6210.

Enhanced Ground Proximity Warning System (EGPWS) The Honeywell MK VI Enhanced Ground Proximity Warning System (EGPWS) consists of an TAWS computer installed in the nose avionics bay and a configuration module mounted as part of the P2 connector backshell. TAWS provides an enhanced capability of reducing accidents cause by controlled flight into terrain. The system achieves this by receiving a variety of aircraft parameters as inputs, then applying alerting algorithms to provide the flight crew with aural messages and visual annunciations and display.

The Terrain Avoidance Warning System (TAWS) unit receives GPS position via the MAU for accurate position determination in conjunction with a regional database. One of three databases may be installed in the system (Atlantic, Americas or Pacific). The databases also contain the locations of all runways longer than 2000 feet (600 m) that have a published instrument approach. A visual depiction of the terrain proximate to the aircraft can be displayed on the PFD and the MFD.

For aircraft with S.O.O. 6388 incorporated, the Enhanced Ground Proximity Warning System (EGPWS) (TAWS) are not installed.

For aircraft with S.O.O. 6388 incorporated, the Enhanced Ground Proximity Warning System (EGPWS) (TAWS) and associated systems are installed by incorporating customer Standard Order Option S.O.O. 6387.